

Family 0.9.tf: land and coast through a published embedding a dataset design note — Google’s AlphaEarth annual 10 m embeddings in place of the raw imagery

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Abstract

Family 0.9.tf is the alternative to family 1.0.tf’s image catalogue: instead of learning a local codec over petabytes of raw optical, radar and LiDAR scenes, it ingests the embedding Google already computed from the same optical and radar imagery, trained against the same LiDAR, L-band radar and climate sources the catalogue lists — the AlphaEarth Foundations Satellite Embedding, version 1: one 64-dimensional unit-length vector per 10 m pixel per year, 2017–2025, for every land surface and shallow water, published under CC BY 4.0, 64 TB a year and 577 TB in all, readable anonymously from a complete mirror. The version number below 1.0 says what it is: a published summary standing in for the raw record, the fastest route to ten-metre land information the programme can take, bought at the price of an annual cadence (73 times coarser than the five-day rule), a closed model that cannot be re-run on new sensors or sub-annual periods, and a record that begins in 2017. The note says what the embedding is and what it was trained on, how it enters family 10’s token without leaking a year’s data into its own anchors, three stores at three resolutions that make it usable at 0.2 TB in the first phase rather than 577 (a 1 km pooled field with a coherence channel, raw 10 m samples at fixed sub-cell positions, and 100 m over regions of interest), the 10 m original referenced by byte range rather than copied, the arithmetic, the one-pass build near the data, the falsifiers that decide it against family 1.0.tf’s own codec, and the decisions left to Chris.

1 Purpose

Family 1.0.tf holds the land’s raw fine record as a catalogue of scenes, and its imagery becomes usable only once a local codec (E-078 §4) has been trained to turn tiles into tokens — the one compute-bound rung on the programme’s ladder, and one whose first build is coastal and small (E-078c). Google’s AlphaEarth Foundations model has already done that job at planetary scale: it reads the same optical and C-band radar imagery the catalogue lists (Sentinel-2, Sentinel-1, Landsat 8 and 9) and was trained to reproduce the rest of it — L-band radar, spaceborne LiDAR, climate and gravity fields, elevation, land-cover labels and geo-linked text — as targets, so those sources shape the embedding without being read when it runs; and its annual output is public. Family 0.9.tf takes that output as a token source, so that the land side of “predict everything” can be tested *now*, against the land targets of family 1.0.tf (forest disturbance, loss year, biomass, burned area, water extent, crop and land cover), and so that the programme can measure whether its own codec is needed for land at all.

The number 0.9 is deliberate. It is below 1.0 because the source is not an observation but a learned summary of observations; the family is a stepping stone, and it is meant to be *compared* with 1.0.tf, not merged into it. If AlphaEarth tokens predict the land targets as well as the programme’s own codec would, the codec is not worth its compute for land at annual cadence; if they do not — because the targets move faster than a year, or because the closed model cannot see NISAR, or because 2017 is too late — the comparison says exactly where 1.0.tf’s raw record earns its cost.

2 What the embedding is

Checked against the paper (Brown et al. 2025, arXiv 2507.22291, v2 2025-09-08), the Earth Engine catalogue entry, Google’s storage read-me and a live listing of both buckets on 2026-09-16.

The model. An “embedding field” model of about 480 M parameters (a 1 B variant was trained; the released layers come from the 480 M one). Its *inputs* are image sequences from Sentinel-2, Sentinel-1 C-

band radar and Landsat 8 and 9. Its training *targets*, which the model learns to reconstruct and therefore encodes without reading them at run time, are ALOS PALSAR-2 L-band radar, GEDI LiDAR, ERA5-Land climate, GRACE gravity, the Copernicus GLO-30 elevation model, NLCD land-cover labels, and Wikipedia articles linked to GBIF species occurrences — more than three billion observations in all, over about 1.1 % of the land surface. No sub-metre imagery is used. The model separates a *support period* (the timestamps of its inputs) from a *valid period* the output summarises, and can summarise, interpolate or extrapolate to a valid period with no observations in it; the released product uses a valid period of one calendar year, and the exact support window of the released layers is not stated.

The output. Per 10 m pixel per year, a 64-dimensional vector of unit length — the vectors lie on the sphere S^{63} , spread by a batch-uniformity objective — which the producer describes as linearly composable, with the dot product between two years at one pixel as the documented change measure (cosine similarity, since the vectors are unit length). Coverage is terrestrial surfaces and shallow water; years 2017–2025 (the 2025 layer was released 2026-03-09; Google commits to annual production with a year’s notice of change, contingent on Landsat and Copernicus continuing).

The files. Cloud-optimised GeoTIFFs of 8192×8192 pixels with 64 bands, in local UTM projections, 120 zone directories per year, about 34,000 tiles a year, 1.9 GB on average. Values are `int8` with nodata `−128` and the documented de-quantisation

$$x = \left(\frac{q}{127.5} \right)^2 \text{sign}(q), \quad (1)$$

so the encoding is $q \approx \text{round}(127.5 \text{sign}(x) \sqrt{|x|})$ — a square-root companding that spends resolution near zero. The producer states the 8-bit form costs negligible accuracy against float32.

Where it lives, measured. `gs://alphaearthfoundations/satellite_embedding/v1/annual/<year>/<zone>/`, with index files (`aef_index.parquet`, 70 MB, updated 2026-02-27); the bucket has been provider-pays since July 2026, so a copy costs the reader nothing. A complete mirror on Source Cooperative (`tge-labs/aef`, AWS Open Data, `s3://us-west-2.opendata.source.coop/tge-labs/aef/v1/annual/`, anonymous S3 and HTTPS) has the same tile counts and byte totals as the origin for every year, plus one VRT per tile; its read-me warns that the COGs are stored bottom-up, so they are read through the VRTs. Earth Engine serves the same layers as floats in $[-1, 1]$ (bands `A00–A63`).

year	tiles	TB
2017	33,148	64.35
2018	33,291	64.15
2019	33,385	64.06
2020	33,206	63.97
2021	33,281	63.91
2022	33,700	63.11
2023	34,151	63.85
2024	34,149	63.90
2025	34,155	65.29
total	$\approx 302,000$	576.6

Table 1: The published embedding, as listed on 2026-09-16 (origin and mirror agree year by year). About 1.4×10^{12} land pixels a year.

Licence. CC BY 4.0, attribution “The AlphaEarth Foundations Satellite Embedding dataset is produced by Google and Google DeepMind”. Every store below carries it; `licence.redistribution` is `attribution` and `distribution` is `public`. The two-track rule of family 1.gf §4 applies, and nothing in this family needs the private track today.

What is not available. Only version 1 and only annual. Sub-annual embeddings (quarterly, monthly, weekly, “down to a 5-day frequency”) exist as Google Maps Platform’s Custom Satellite Embeddings, in private preview with no public licence; an academic programme offers samples through Earth Engine, applications closing 2026-10-15. A granted sample is usable here under the programme’s terms as a private-track store, `aef_custom` (family 1.gf §4), never republished — the one place this family would use that

track. The model and training code are closed: no new sensor (NISAR), no new year on demand, no re-run over a chosen period. Reported weaknesses: transfer across regions and years is limited (an irrigation study reproduced 0.93–0.95 accuracy locally and about 0.3 when moved to another region), short events are averaged away by the annual window, and the model receives no coordinates, so a location-dependent task needed far more labels than a coordinate-aware baseline.

3 Grid, time axis and token

The embedding is a 10 m raster in UTM; the family’s grids are latitude and longitude. The stores below resample by *pooling*, never by interpolating a single value, and the 10 m original is referenced, not copied (§4.4), so nothing is resampled at storage time except what is declared as a pooled field with its own footprint.

Time. A year- Y embedding summarises 1 January to 31 December of Y : $\log_2 f_t = \log_2(365/5) = +6.2$. It is filed under the **first family-10 bin that begins after 31 December Y** — not the bin containing that day, because the reader includes the anchor’s own bin (family 10.1 §7: candidates are “at or before the anchor’s pentad”), and a year-end bin would hand an anchor in the last days of Y its own year’s summary. With that filing the one-sided rule does the rest: an anchor in any bin of year Y sees the embedding of year $Y - 1$ and earlier, never Y ’s own. Δt is measured from the end of the anchor’s bin to 31 December Y , 0 to about 370 days for the most recent usable year. This is the strict rule and the pick; the relaxed alternative — letting an anchor in Y read Y ’s embedding when the target lies after Y — is a leak of up to a year of future imagery into the input and is not offered. One caveat the rule cannot close: the producer does not state the support window of the released layers, so if year Y ’s embedding was computed from observations extending past 31 December Y , even this filing admits them; the registry records the assumption.

Footprint. A raw 10 m vector carries $\log_2 f_p = \log_2(0.01/27.83) = -11.4$; a pooled $1/1200^\circ$ cell (0.093 km) -8.2 ; a pooled $1/120^\circ$ cell (0.93 km) -4.9 . All are inside the registry’s ± 12 clamp.

The token. The 64 components are the token’s value vector with $C = 64$ (plus one coherence channel for pooled stores, §4.1), raw units (unit-length components in $[-1, 1]$), mask all-true where the pixel is not nodata; source embedding “AlphaEarth v1”; offsets and footprints as everywhere else. Nothing is z-scored: the components are already on a sphere, and the loader standardises from training years if it wants to.

4 The stores

Three stores at three resolutions, and one reference. Bytes assume int8 components (the producer’s own precision) plus one uint8 coherence channel, so a pooled pixel costs 65 bytes and a raw 10 m vector 64.

4.1 aef1k — the pooled 1 km field with coherence, tier G sharded

On a $1/120^\circ$ grid (≈ 0.93 km, 43200×21600), each cell’s value is the **mean of the unit vectors** of its $\approx 100 \times 100$ valid 10 m pixels, **renormalised** to unit length, stored as int8 under the producer’s companding; and a 65th channel, the **coherence**: the length of the un-normalised mean, in $[0, 1]$, 1 where the square kilometre is one thing and near 0 where it is many. The coherence is what the mean throws away, kept as a number, so that a 1 km token says both “what” and “how uniformly”. One frame per year ($F = 1$), filed under the bin of 31 December. Land is $\approx 29\%$ of the grid, ≈ 270 M valid cells: $270 \text{ M} \times 65 \times 1.2 \approx 21$ GB per year, ≈ 0.2 TB for 2017–2025 (0.25 TB if the high-latitude land the 29 % undercounts is included).

4.2 aef_dots — raw 10 m vectors at fixed sub-cell positions, tier P

Pooling loses the fine structure the embedding was made for. **aef_dots** keeps sixteen *unpooled* 10 m vectors per 0.25° land cell per year at fixed sub-cell positions — the cone’s own sunflower offsets scaled into the cell, the same sixteen positions every year so that a position’s time series is a time series — as tier-P rows: **lat**, **lon** of the pixel centre, **time_s** = 31 December Y , **platform** = the hashed tile identifier, **qc** = 1, **values** the 64 components as float16 ($C = 64$; int8 would need a schema change and is not worth it at

this size). About $300\text{ k land cells} \times 9\text{ years} \times 16 = 43\text{ M rows} \times 155\text{ bytes} \approx \mathbf{7\text{ GB}}$. This is the store a consumer reads to ask what 10 m detail adds over the kilometre mean, for 7 GB.

4.3 aef100r — 100 m over regions of interest, tier G sharded

The same pooling at $1/1200^\circ$ ($\approx 93\text{ m}$) over declared regions: first the deforestation frontiers (the Amazon arc, the Congo basin, insular Southeast Asia, $\approx 5\text{ M km}^2$), then the $\pm 100\text{ km}$ coastal band of family 1.0.tf. Regions: $5 \times 10^8\text{ cells} \times 65 \times 1.2 \approx 39\text{ GB}$ per year, $\approx 0.35\text{ TB}$ for nine years; the coastal band (taken as 30 % of land, $\approx 8 \times 10^9\text{ cells}$ at this grid) would be $\approx 0.6\text{ TB}$ per year, $\approx 5.7\text{ TB}$ in all, and is phase C. The region list is a registry field, and a new region is a new build of that region alone.

4.4 aef10 — the original, referenced by byte range

The 10 m tiles are not copied. The registry carries a group of layout "external-cog": the mirror's base URL, the per-tile index (`aef_index.parquet`, 70 MB, re-hosted on the Hub with its sha256), the projection per zone, the companding formula, and the bottom-up orientation flag. A read is a COG-internal byte range on an anonymous bucket, the same operation the family-7 globe layer performs on the Hub, and the reader returns raw 10 m vectors for a cone's dots on demand. This is the family's honest position on 577 TB: it is readable, it is not ours to host, and the day the mirror goes away the three copied stores above are what remains — which is why they exist.

store	tier	content	footprint ($\log_2 f_p, \log_2 f_t$)	rows or cells	stored	phase
aef1k	G sharded	64 pooled + 1 coherence, int8/uint8, $1/120^\circ$, one frame per year	$(-4.9, +6.2)$	270 M land cells $\times 9$	$\approx 0.2\text{ TB}$	A
aef.dots	P	16 raw 10 m vectors per 0.25° land cell per year, float16	$(-11.4, +6.2)$	43 M rows	7 GB	A
aef100r	G sharded	pooled 100 m over regions of interest; the coastal band later	$(-8.2, +6.2)$	$5 \times 10^8\text{ cells} \times 9$	$\approx 0.35\text{ TB}$ (regions); $\approx 5.7\text{ TB}$ (band)	B / C
aef10	external	the mirror's COGs by byte range; index and formula in the registry	$(-11.4, +6.2)$	$1.4 \times 10^{12}\text{ px} \times 9$	0 (577 TB referenced)	A

Table 2: The four groups of family 0.9.tf.

5 Missing values

Nodata (-128 in the source) is NaN in every store and never a zero vector; a pooled cell with fewer than 10 % valid pixels is NaN with its coherence NaN, and the valid fraction is written to the shard index so a consumer can raise the threshold. Water beyond the shallow zone, the poles above the product's coverage and the years before 2017 are absent, and absent is what the token says ($n_R = 0$).

6 Design decisions

Pool by the mean of unit vectors, and keep its length. The producer says the vectors are linearly composable and that the dot product is the similarity; the mean of unit vectors is then the natural pooled representative, its length the natural measure of agreement. Any other pooling (a medoid, a random pixel, a principal direction) discards either the "what" or the "how uniform". The dots store keeps raw samples beside the mean so the pooling choice is testable rather than assumed.

Strict availability by year end. The one rule that must not bend: an embedding of year Y is not visible to an anchor inside Y . The cost is a lag of up to a year on the most recent land state, which is real and is the price of an annual product; the alternative is a leak that would make every land read-out look better than it is.

Build once, next to the data. Producing `aef1k` and `aef_dots` means reading all 577 TB once. On a rented box at 2 Gbps that is 27 days and the transfer risk of the family-10.1 `slatrack` night; on a spot instance in the mirror’s own region (us-west-2, in-region S3 reads free, anonymous) it is about five days at 10 Gbps and \$50–150 of compute, writing year-parts to the Hub as it goes with `done.json` last, the family-10 lane discipline. Pick: in-region, one lane per year, the first year measured before the other eight are dispatched.

Reference the original rather than copy it. 577 TB is about seventy times the largest store any family plans and is CC BY 4.0 on two public buckets; copying it would cost more than every other family together and add nothing. The referenced group is the honest shape: the registry says where the bytes are and how to read them, and the copied stores are what the programme guarantees.

Why not stop at 1 km. A kilometre is the family-7 resolution’s neighbour, not the point. The dots store (7 GB) and the regional 100 m store are what let the programme ask whether the land targets need 10 m inputs, at a cost that is a rounding error next to the raw families.

What this family cannot do, said plainly. It cannot see a five-day event, cannot be extended to NISAR or to any sensor Google did not train on, cannot go back before 2017, and cannot be re-run over a chosen support window; if Google changes or stops the product, the copied stores freeze at the last year. Those are the reasons family 1.0.tf exists, and the two are compared, not merged.

7 Validation

Every store as in family 10.1 (hashes, N from file sizes, bin identity, CSR index) and family 1.gf (shards decompressing to declared shapes); plus three checks specific to this source, each on every tile read: the de-quantised vectors are unit length to within 10^{-2} — the int8 companding itself perturbs the length by up to 8.5×10^{-3} , so a tighter tolerance would reject correct data, and a wrong companding or a wrong orientation breaks it by far more (the encoder must also clip to ± 127 , since $|x| = 1$ maps to 128); the bottom-up orientation is asserted by comparing a tile’s corner against the index polygon; and the per-year tile count and byte total equal the listing in the table above, so a truncated year is refused by name.

8 Limitations

Annual, closed, 2017 onwards, terrestrial and shallow water only, and a learned summary whose failure modes are the producer’s (transfer across regions and years, short events averaged out). The support window of the released layers is not published, so “what observations went into year Y ” is not knowable pixel by pixel. The regional list for `aef100r` is a choice, and a region not on it has only the 1 km field and the dots. Nothing is dispatched on an unmeasured size: the first year’s build measures the pooling throughput, the coherence distribution and the dots’ bytes before the rest.

9 Next steps: the ladder, the probe and the falsifiers

Phase A. `aef1k`, `aef_dots`, the `aef10` reference group and the registry; ≈ 0.2 TB; one in-region lane per year. **Phase B.** `aef100r` over the deforestation frontiers. **Phase C.** the coastal band at 100 m, once E-078c’s coastal cells exist to compare against.

The probe. One UTM zone of one year (about 280 tiles, 0.5 TB) through the real pooling code: throughput, the valid fraction, the coherence histogram, the orientation check, and the bytes of the resulting shards — the numbers that replace this note’s before the nine-year build. The Hugging Face sample (Major-TOM/Core-AlphaEarth-Embeddings, 62.5 k grid cells, 2017–2023) is the unit-test fixture.

Three falsifiers, three seeds each.

1. **Against the programme’s own codec.** On the land pixels of E-078c’s ≈ 300 coastal cells, tokens from `aef_dots` and `aef1k` against the local codec’s cached tokens, on the *annual* land targets in

the band (loss year, land cover, biomass) — not on E-078c’s own pentad ocean targets, which the embedding does not cover and whose cadence it cannot match. If AlphaEarth matches or beats the codec there, the codec is not needed for land at annual cadence and E-078c’s scope narrows to what the embedding cannot see.

2. **Against the coarse tensor, on the land targets.** Next year’s forest alerts, loss year, burned area and water extent (family 1.0.tf’s target stores) predicted from family 7.2 alone versus family 7.2 plus this family’s tokens, at the targets’ own footprints. Parity means 10 m information does not reach those targets through an annual summary.
3. **What the annual cadence costs.** The same targets from this family plus family 1.0.tf’s daily fields (temperature, reflectance, snow, fire) versus this family alone. The gap is the value of the five-day rule on land, measured rather than argued, and it is the number that decides how much of family 1.0.tf’s image tier to build.

10 Decisions for Chris

1. **The name:** 0.9.tf as given, the number reading as “an embedding in place of the raw record”. Pick: as given.
2. **Strict availability by year end.** Pick: strict.
3. **Pooling:** renormalised mean plus coherence. Pick: yes, with the dots store as the check.
4. **Where to build:** a spot instance in the mirror’s region rather than a rented box. Pick: in-region.
5. **The regional list for 100 m:** the three deforestation frontiers first. Pick: yes.
6. **Custom Satellite Embeddings:** the academic programme for sub-annual embeddings closes 2026-10-15; applying costs a form and would make falsifier 3 answerable at five-day cadence on the same model. Pick: apply; a granted sample becomes the private-track store `aef_custom`.

Terms used above, in one line each

AlphaEarth Foundations (AEF) Google DeepMind’s model that turns Sentinel-1, Sentinel-2 and Landsat image sequences into one 64-number description of each 10 m of land per year, trained to reproduce radar, LiDAR, climate and land-cover targets; **Satellite Embedding V1** the published output. **Embedding** a learned vector summarising what was observed. **Unit-length** / S^{63} vectors of length one on a 63-dimensional sphere, so their dot product is a similarity. **int8 companding** storing a value in one byte on a square-root scale. **COG** / **VRT** / **UTM** cloud-optimised GeoTIFF (readable by byte range), a GDAL virtual-raster wrapper, and the projected map zones the tiles are in. **Tier G** / **P** family 10’s gridded and point storage tiers. **Coherence** here, the length of the mean of unit vectors in a cell: 1 if all agree, 0 if they cancel. **Sentinel-1/2**, **Landsat** the model’s inputs: C-band radar and optical imagery from ESA and optical from NASA/USGS. **PALSAR-2**, **GEDI**, **ERA5-Land**, **GRACE**, **GLO-30**, **NLCD**, **GBIF** its training targets: L-band radar from JAXA, spaceborne LiDAR, a land climate reanalysis, satellite gravity, an elevation model, the US land-cover map, and species-occurrence records. **SST** sea-surface temperature. **NaN** not-a-number, the stored value for “not measured”. **CSR** compressed sparse row, the bin index of a point store. **Earth Engine** Google’s hosted geospatial platform, which needs an account. **S3** / **us-west-2** / **spot instance** Amazon’s object storage, its Oregon region where the mirror lives, and a discounted pre-emptible virtual machine. **sha256** the file hash every store carries. **NISAR** the NASA-ISRO L-band radar launched 2025, not among the inputs. **E-078** / **E-078c** the multi-granularity design and its first coastal tile-codec experiment. **Source Cooperative** / **AWS Open Data** / **GCS** the mirror’s host, the sponsorship it runs under, and Google Cloud Storage. **Major-TOM** an open Earth-observation dataset project whose Hugging Face sample of the embeddings serves as a test fixture. **Public** / **private track** a store anyone may read from the public repository, or one held for this project’s own use because its licence forbids redistribution (family 1.gf §4).

References

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